



Name of the module: FD_QEC
Location in GIT repository: `vsps_common\vsps-lib\fd_qec`

Overview:

This module provides kernels on that performs the frequency domain quadrature error correction.

Assumptions:

The following assumptions were made:

- Input buffers precision are *Fix Point Half Precision*
- Output buffer precision is *Fix Point Half Precision*
- The cases treated correspond to the requirements
- Values for the stream size are given by the integrator

Type of code:

- VCPU assembly functions:
 - `fd_qec`

Assembly Functions API:

```
void fd_qec( cfixed16_t *out_p,  
            cfixed16_t *inp_p0,  
            cfixed16_t *inp_p1,  
            cfixed16_t *inp_p2,  
            uint32_t  size);
```

where

<code>out_p</code>	– output pointer
<code>inp_p0</code>	– input pointer for the first block of data
<code>inp_p1</code>	– input pointer for the second block of data
<code>inp_p2</code>	– input pointer for the weights
<code>size</code>	– buffer size



Implementation plan

We will use following formula to conduct the I/Q correction in frequency domain:

$$Z_k = K_1 Y_k + K_2 Y_{-k}^*$$

with $K_1 = \frac{1 + \alpha e^{-j\phi}}{2}$ and $K_2 = \frac{1 - \alpha e^{j\phi}}{2}$, where α and ϕ represent the gain and phase imbalance.

The kernel assumes that Y_k , K_1 , K_2 and Y_{-k}^* are pre-computed and performs the multiplication and accumulation.

Observations:

Input sample size is assumed to be `fft_size`.
Output is always a multiple of line.
Does not operate in place

Memory management

Output is 128 DMEM lines for `fft_size = 4096`; 64 DMEM lines for `fft_size = 2048`, 32 DMEM lines for `fft_size = 1024`; 16 DMEM lines for `fft_size = 512`

Reference

<https://www.nutq.com/blog/rf-imperfection-and-compensation-part-1-effects-iq-imbalance-and-compensation-receiver>